

MAT31-3

Information Theory & Statistical Learning Foundations

Albert School — 38 hours

Preface

This course is a confluence of two rivers you have already been swimming in.

The first is **analysis**. In **Foundations** (MAT11-1) you met sequences, limits, derivatives, integrals, and — crucially — Taylor approximations, where an infinite sum $\sum a_n x^n$ was used to replace a complicated function by a polynomial near a point. You used those sums freely: you differentiated $e^x = \sum x^n/n!$ term by term, you integrated, you swapped limits and sums almost without comment. The first half of this course goes back and asks the question that was politely deferred: **when are these manipulations actually legal?** The answer is the theory of **uniform convergence**, and it is not a pedantic detail — it is exactly the discipline that makes approximation and estimation theory rigorous, and it culminates in a theorem (Banach’s) that proves **value iteration**, the workhorse algorithm of reinforcement learning, converges.

The second river is **probability and statistics**. In MAT21-2 you built discrete probability and met **maximum likelihood estimation**; in MAT31-2 (running alongside this course) you extended to continuous variables, the Central Limit Theorem, and a first glimpse of the **Bellman equation**. The second half of this course forges the **information-theoretic** tools that turn those probabilistic foundations into the language of machine learning: **entropy** (how much uncertainty a random variable carries), **mutual information** (how much one variable tells you about another), **Fisher information** (how much a data point tells you about a parameter), and **KL divergence** (how far one distribution is from another). The payoff is a single, clarifying identity: minimizing the **cross-entropy loss** that trains a neural network **is** maximizing likelihood.

The two halves are not bolted together by accident. The information-theoretic proofs — that mutual information is non-negative, that the Cramér–Rao bound holds, that cross-entropy is minimized at the truth — all lean on convexity, on interchanging limits with integrals, and on the convergence discipline of the first half. Convergence theory is not a warm-up you leave behind; it is the scaffolding the second half is built on.

Keep one slogan in view throughout: **information is reduction of uncertainty, and uncertainty is measured in bits**. Everything — entropy, mutual information, the link to learning — is a variation on that single theme.

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1 Power series

You have already used power series without ceremony. The exponential, sine, and logarithm of MAT11-1 were all secretly infinite sums:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, \quad \ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}.$$

Each of these is valid only for certain x . The exponential series converges for **every** real x ; the logarithm series, as we will see, only for $|x| < 1$. The first question about any power series is therefore: **for which x does it converge?** The answer always has the same shape — an interval centred at the origin — and its half-width is the single most important number attached to the series.

1.1 The radius of convergence

Definition 1 A **power series** centred at 0 is a series of the form

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots,$$

with **coefficients** $a_n \in \mathbb{R}$. There is a number $R \in [0, +\infty]$, the **radius of convergence**, such that the series

- converges **absolutely** whenever $|x| < R$, and
- **diverges** whenever $|x| > R$.

The open interval $(-R, R)$ is the **interval of convergence**. At the two endpoints $x = \pm R$ anything can happen, and each must be examined separately.

The dichotomy is genuinely clean: inside the radius the terms $a_n x^n$ shrink fast enough to sum; outside, they do not even tend to zero. The only subtlety lives exactly on the boundary circle $|x| = R$.

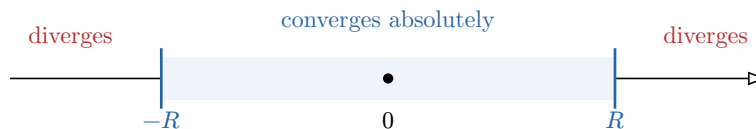


Figure 1: The convergence set of a power series is always an interval centred at 0. Inside the radius R the series converges absolutely; outside it diverges; the two endpoints $\pm R$ are undecided by R alone and need a separate test.

1.2 Computing R : the ratio and root tests

In practice R is read off the coefficients. Both tests come straight from the convergence tests for numerical series applied to $\sum |a_n x^n|$.

Theorem 2 (Ratio and root tests for the radius) For the power series $\sum a_n x^n$, provided the relevant limit exists,

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \quad (\text{ratio test}), \quad \frac{1}{R} = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} \quad (\text{root test}),$$

with the conventions $1/0 = +\infty$ (the series converges everywhere) and $1/(+\infty) = 0$ (it converges only at $x = 0$).

Worked example — three radii by the ratio test.

- **Exponential** $\sum x^n/n!$: $|a_{n+1}/a_n| = n!/(n+1)! = 1/(n+1) \rightarrow 0$, so $1/R = 0$ and $R = +\infty$ — convergence for every x , as promised.
- **Logarithm** $\sum_{n \geq 1} x^n/n$: $|a_{n+1}/a_n| = n/(n+1) \rightarrow 1$, so $R = 1$. The series converges on $(-1, 1)$ and diverges for $|x| > 1$.
- **Geometric** $\sum x^n$: $a_n = 1$ for all n , so $1/R = 1$ and $R = 1$; indeed $\sum x^n = 1/(1-x)$ exactly when $|x| < 1$.

Worked example — the root test earns its keep. For $\sum 2^n x^{2n} = \sum (2x^2)^n$ the coefficients alternate between 0 and powers of 2, so the **ratio** a_{n+1}/a_n is ill-defined. The root test still works on the series in the variable $u = x^2$: it converges when $2x^2 < 1$, i.e. $|x| < 1/\sqrt{2}$, giving $R = 1/\sqrt{2}$. When coefficients vanish periodically, reach for the root test.

Common pitfall The radius R says **nothing** about the endpoints. The three series

$$\sum x^n, \quad \sum \frac{x^n}{n}, \quad \sum \frac{x^n}{n^2}$$

all have $R = 1$, yet at $x = 1$ the first diverges, the second diverges (it is the harmonic series), and the third converges; at $x = -1$ the first diverges while the second converges (alternating harmonic). Same radius, three different endpoint behaviours. Always test $x = R$ and $x = -R$ by hand.

1.3 Differentiating and integrating term by term

The reason power series are so useful is that, **strictly inside** the radius, they behave like very long polynomials: you may differentiate and integrate them one term at a time.

Theorem 3 (Term-by-term differentiation and integration) Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ have radius of convergence $R > 0$. Then f is infinitely differentiable on $(-R, R)$, and on that interval

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}, \quad \int_0^x f(t) dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}.$$

Both resulting series have the **same** radius of convergence R .

That the radius is preserved is what makes the operation safe to iterate: you may differentiate again, and again, never leaving the interval where everything converges.

Worked example — deriving a series by integrating a known one. Start from the geometric series $1/(1-t) = \sum_{n=0}^{\infty} t^n$, valid for $|t| < 1$. Replace t by $-t$ and integrate from 0 to x :

$$\ln(1+x) = \int_0^x \frac{1}{1+t} dt = \int_0^x \sum_{n=0}^{\infty} (-1)^n t^n dt = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$$

Term-by-term integration **manufactured** the logarithm series from the geometric one — and the theorem guarantees it is valid on the same interval $|x| < 1$.

Common pitfall Term-by-term operations are licensed **only inside** the open interval $(-R, R)$, never automatically at the endpoints, and never outside. The differentiated series has the same radius but possibly **worse** endpoint behaviour: differentiating $\sum x^n/n^2$ (which converges at $x = 1$) gives $\sum x^{n-1}/n$, which diverges at $x = 1$. Inside the radius you are safe; on the boundary you are on your own.

Exercises

1. Find the radius of convergence of $\sum n^2 x^n$, $\sum x^n/n!$, and $\sum n! x^n$.
2. Determine R for $\sum x^n/(n 3^n)$ and then decide convergence at both endpoints $x = 3$ and $x = -3$.
3. Starting from $1/(1-x) = \sum x^n$, differentiate term by term to obtain a closed form for $\sum_{n \geq 1} n x^{n-1}$ on $(-1, 1)$.
4. Use the root test to find the radius of $\sum (n/(n+1))^{n^2} x^n$.

2 Pointwise and uniform convergence

A power series is a special case of something more general: a **sequence of functions** f_n that we hope tends to a limit function f . The partial sums $S_N(x) = \sum_{n=0}^N a_n x^n$ are exactly such a sequence. But “ f_n tends to f ” can mean two genuinely different things, and the gap between them is the single most important distinction in this half of the course. Get it wrong and you will “prove” false statements — a continuous limit that is secretly discontinuous, an integral that jumps.

2.1 A motivating example: the migrating bump

Consider $f_n(x) = x^n$ on the interval $[0, 1]$. Fix any single point x and watch the sequence of **numbers** $f_n(x)$:

- if $0 \leq x < 1$, then $x^n \rightarrow 0$;
- if $x = 1$, then $x^n = 1$ for all n , so the limit is 1.

So **for each fixed x** the sequence converges, to the function

$$f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x = 1 \end{cases}.$$

Every f_n is continuous, yet the limit f has a jump at $x = 1$. Something has gone wrong with the naive hope that “limit of continuous is continuous”. The diagnosis is that the convergence, while holding at each point, is not **uniform**: near $x = 1$ the graphs of f_n stay stubbornly far from f no matter how large n is.

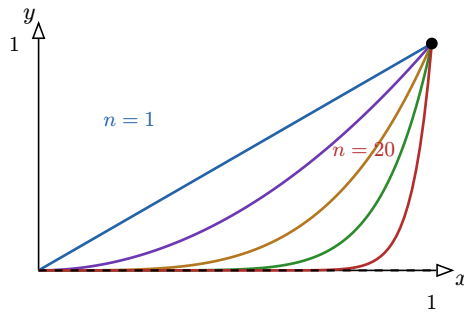


Figure 2: The functions $f_n(x) = x^n$ on $[0, 1]$. At every fixed $x < 1$ the height collapses to 0 as n grows, so $f_n \rightarrow f$ **pointwise** with f the dashed line (and $f(1) = 1$, the dot). But the “knee” of each curve never flattens: $\sup_x |f_n(x) - f(x)| = 1$ for every n , so the convergence is **not uniform**.

2.2 The two definitions

Definition 4 Let $f_n : D \rightarrow \mathbb{R}$ be functions on a set D , and $f : D \rightarrow \mathbb{R}$.

- $f_n \rightarrow f$ **pointwise** on D if for every fixed $x \in D$ the numerical sequence $f_n(x)$ converges to $f(x)$: for each x and each $\varepsilon > 0$ there is N (which **may depend on x**) with $|f_n(x) - f(x)| < \varepsilon$ for $n \geq N$.
- $f_n \rightarrow f$ **uniformly** on D if

$$\sup_{x \in D} |f_n(x) - f(x)| \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Equivalently: for each $\varepsilon > 0$ there is a **single N — the same for all x —** with $|f_n(x) - f(x)| < \varepsilon$ for all $n \geq N$ and all x .

The entire difference is the order of the quantifiers. Pointwise: **for each x , there is an N** . Uniform: **there is an N , good for every x at once**. The quantity $\|f_n - f\|_\infty = \sup_x |f_n(x) - f(x)|$ — the worst-case error over the whole domain — is the **supremum criterion**, and uniform convergence is precisely the statement that this single number tends to zero.

Proposition 5 Uniform convergence implies pointwise convergence. The converse is false.

Proof. If the worst-case error $\sup_x |f_n(x) - f(x)|$ tends to 0, then in particular at each fixed x the error $|f_n(x) - f(x)|$, being no larger, also tends to 0 — that is pointwise convergence. The migrating bump $f_n(x) = x^n$ shows the converse fails: it converges pointwise but $\sup_{x \in [0,1)} x^n = 1 \not\rightarrow 0$. $\square$

Common pitfall The single most common error is to compute the limit at each point, find it exists, and conclude “uniform convergence”. **Pointwise convergence is automatic bookkeeping; uniform convergence is a genuine claim about the supremum.** Always finish the job: compute $\sup_x |f_n(x) - f(x)|$ and check that this **number** goes to zero. To find the supremum, maximize $|f_n(x) - f(x)|$ over x as a function of x (often by differentiating in x) for each fixed n .

2.3 Why uniformity matters: continuity is preserved

The reward for uniform convergence is that the limit inherits the good behaviour of the f_n .

Theorem 6 (Uniform limit of continuous functions) If each f_n is continuous on D and $f_n \rightarrow f$ **uniformly** on D , then f is continuous on D .

Proof. Fix x_0 and $\varepsilon > 0$. By uniform convergence pick N with $\sup_x |f_N(x) - f(x)| < \varepsilon/3$. Since f_N is continuous at x_0 , choose δ so that $|x - x_0| < \delta$ gives $|f_N(x) - f_N(x_0)| < \varepsilon/3$. Then for such x , the **triangle inequality** (MAT11-1) splits the error through f_N :

$$|f(x) - f(x_0)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(x_0)| + |f_N(x_0) - f(x_0)| < \varepsilon.$$

This is the celebrated “ $\varepsilon/3$ argument”: uniformity is exactly what lets the **same** N control the first and third terms regardless of where x sits. $\square$

Remark The migrating bump is the perfect cautionary tale: continuous f_n , but a **discontinuous** pointwise limit. The theorem is not violated — its hypothesis (uniform convergence) simply fails. Read contrapositively, the theorem is a powerful **disproof tool**: if the f_n are continuous but the limit is not, the convergence **cannot** have been uniform. No supremum computation needed.

Exercises

5. Show $f_n(x) = x/n$ converges to 0 uniformly on $[0, 1]$ but only pointwise on all of $\mathbb{R}$.
6. For $f_n(x) = nx(1-x)^n$ on $[0, 1]$, find the pointwise limit, then compute $\sup_x |f_n(x)|$ by maximizing in x and decide whether the convergence is uniform.
7. Show $f_n(x) = x^n(1-x)$ on $[0, 1]$ converges uniformly to 0. (Maximize over x for fixed n .)
8. Use the continuity theorem to argue, without any supremum computation, that $f_n(x) = \arctan(nx)$ does **not** converge uniformly on $\mathbb{R}$.

3 Interchange theorems

Pointwise and uniform convergence are not abstractions for their own sake: they are the gatekeepers of the manipulations you have been performing since MAT11-1. “The integral of the limit is the limit of the integrals” and “the derivative of the limit is the limit of the derivatives” are **not free**. They are theorems, each with hypotheses, and each fails in instructive ways when the hypotheses are dropped. Treating limit–integral and limit–derivative exchanges as **theorem-governed operations**, never reflexes, is the central habit of this chapter.

3.1 When may a limit pass through an integral?

Worked example — the integral that does not follow the limit. On $[0, 1]$ let $g_n(x) = n$ on $(0, 1/n)$ and 0 elsewhere — a tall thin spike of area 1. For each fixed $x > 0$ eventually $1/n < x$, so $g_n(x) \rightarrow 0$; and $g_n(0) = 0$. Thus $g_n \rightarrow 0$ pointwise. Yet

$$\int_0^1 g_n(x) dx = 1 \quad \text{for all } n, \quad \text{while} \quad \int_0^1 \lim_n g_n(x) dx = \int_0^1 0 dx = 0.$$

The limit and the integral **cannot be swapped here** — the answers are 1 and 0. The spike carries its area off “to infinity” in height while shrinking in width, and pointwise convergence is blind to it. The cure is uniform convergence, which forbids such escaping mass: here $\sup_x |g_n(x)| = n \rightarrow \infty$, the opposite of uniform.

Theorem 7 (Interchange of limit and integral) If $f_n \rightarrow f$ **uniformly** on a bounded interval $[a, b]$ and each f_n is continuous, then f is continuous (hence integrable) and

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx = \int_a^b \lim_{n \rightarrow \infty} f_n(x) dx = \int_a^b f(x) dx.$$

Proof. The error between the two sides is controlled by the worst-case error:

$$\left| \int_a^b f_n - \int_a^b f \right| \leq \int_a^b |f_n - f| dx \leq (b - a) \sup_{x \in [a, b]} |f_n(x) - f(x)|.$$

Uniform convergence sends the right-hand side to 0. The bounded length $(b - a)$ is essential — this is why the theorem is stated on a bounded interval. $\square$

3.2 When may a limit pass through a derivative?

Differentiation is more delicate than integration, because the derivative is sensitive to **small wiggles** that integration averages away. Here uniform convergence of the f_n themselves is **not enough** — it is the **derivatives** that must converge uniformly.

Theorem 8 (Interchange of limit and derivative) Suppose on $[a, b]$ each f_n is continuously differentiable, the sequence $f_n \rightarrow f$ converges pointwise, and the **derivatives** $f_{n'} \rightarrow g$ converge **uniformly**. Then f is differentiable and

$$f'(x) = \lim_{n \rightarrow \infty} f_{n'}(x) = g(x).$$

Common pitfall Uniform convergence of f_n does **not** license differentiating term by term. Take $f_n(x) = \sin(nx)/\sqrt{n}$ on $[0, 2\pi]$. Then $\sup_x |f_n(x)| \leq 1/\sqrt{n} \rightarrow 0$, so $f_n \rightarrow 0$ **uniformly** — as nicely as one could want. But

$$f_{n'}(x) = \sqrt{n} \cos(nx),$$

whose amplitude $\sqrt{n} \rightarrow \infty$: the derivatives blow up and converge to nothing. The limit of the functions is 0 with derivative 0, but $\lim f_{n'}$ does not exist. The hypothesis that fails is the one on the **derivatives**, exactly the one the theorem insists upon.

Remark The asymmetry has a clean intuition. Integration **smooths**: it sums up area, so a uniformly small function contributes uniformly small area. Differentiation **sharpens**: it reads off slopes, and a function can be uniformly tiny in value while oscillating with enormous slope (the $\sin(nx)/\sqrt{n}$ example squeezes the graph in height while cramming in ever-faster wiggles). This is why the derivative theorem must put its uniformity hypothesis on $f_{n'}$, not on f_n .

Remark (back to power series) These theorems are what retroactively justified Chapter 1. Inside its radius a power series converges uniformly on every closed sub-interval $[-r, r]$ with $r < R$, and so does the differentiated series. That uniform convergence is exactly the hypothesis the two interchange theorems require — which is **why** term-by-term differenti-

ation and integration are legal inside the radius and why the operation cannot be pushed to the boundary.

Exercises

9. Verify directly that limit and integral may be swapped for $f_n(x) = x^n$ on $[0, 1/2]$, computing both $\lim \int f_n$ and $\int \lim f_n$.
10. For the spike g_n of the worked example, confirm $g_n \rightarrow 0$ pointwise and that $\int_0^1 g_n = 1$; identify precisely which hypothesis of the integral interchange theorem fails.
11. Let $f_n(x) = \cos(nx)/n^2$ on $[0, \pi]$. Show $f_n \rightarrow 0$ uniformly, that $f_{n'}$ converges uniformly, and that the derivative interchange is therefore valid.
12. Explain why the bounded-interval hypothesis matters: find functions $f_n \rightarrow 0$ uniformly on $[0, \infty)$ with $\int_0^\infty f_n = 1$ for all n .

4 Fixed-point arguments and value iteration

The convergence theory now pays a spectacular dividend. There is a single theorem — **Banach's fixed-point theorem** — that simultaneously proves a solution to an equation **exists**, is **unique**, and is **computed** by simple iteration with a **guaranteed error bound**. We then apply it to the central algorithm of reinforcement learning: **value iteration** on a Markov decision process, the algorithmic engine behind the Bellman equation you met as a one-step identity in MAT31-2.

4.1 Contractions

Worked example — halving the distance to a fixed point. Consider $T(x) = x/2 + 3$ on $\mathbb{R}$. Iterating from any x_0 , the distance to the point $x^* = 6$ (which satisfies $T(6) = 6$) **halves** every step: $|T(x) - 6| = |x/2 + 3 - 6| = \frac{1}{2}|x - 6|$. From $x_0 = 0$: $0 \rightarrow 3 \rightarrow 4.5 \rightarrow 5.25 \rightarrow 5.625 \rightarrow \dots \rightarrow 6$. The factor $1/2 < 1$ is what drives the iteration home. A map that shrinks distances by a fixed factor below 1 is called a contraction.

Definition 9 Let $(E, \|\cdot\|)$ be a complete normed space (a normed vector space in which every Cauchy sequence converges — $\mathbb{R}^n$ with any norm is complete). A map $T : E \rightarrow E$ is a **contraction** with **factor** $\gamma \in [0, 1)$ if

$$\|T(x) - T(y)\| \leq \gamma \|x - y\| \quad \text{for all } x, y \in E.$$

The condition $\gamma < 1$ is everything. With $\gamma = 1$ (an **isometry**, e.g. a rotation) points keep their distance and need not converge; with $\gamma > 1$ the map expands and iterates fly apart.

4.2 Banach's theorem

Theorem 10 (Banach fixed-point theorem) A contraction T with factor γ on a complete normed space has a **unique** fixed point x^* (the unique solution of $T(x^*) = x^*$). For **any** starting point x_0 , the iterates $x_{k+1} = T(x_k)$ converge to x^* , with the **geometric error bound**

$$\|x_k - x^*\| \leq \gamma^k \|x_0 - x^*\|.$$

Proof (sketch). Successive iterates contract: $\|x_{k+1} - x_k\| = \|T(x_k) - T(x_{k-1})\| \leq \gamma \|x_k - x_{k-1}\|$, so $\|x_{k+1} - x_k\| \leq \gamma^k \|x_1 - x_0\|$. The gaps are dominated by a convergent geometric series, making (x_k) a Cauchy sequence; completeness gives a limit x^* , and continuity of T (a contraction is continuous) gives $T(x^*) = x^*$. Uniqueness: if $T(x^*) = x^*$ and $T(y^*) = y^*$ then $\|x^* - y^*\| = \|T(x^*) - T(y^*)\| \leq \gamma \|x^* - y^*\|$, and since $\gamma < 1$ this forces $\|x^* - y^*\| = 0$. The error bound follows from $\|x_k - x^*\| = \|T(x_{k-1}) - T(x^*)\| \leq \gamma \|x_{k-1} - x^*\|$, iterated k times. $\square$

Remark The error bound $\gamma^k \|x_0 - x^*\|$ is a precise **convergence rate**: each iteration multiplies the error by γ , so the number of correct digits grows linearly in k . To shrink the error below ε you need about $k \approx \log(\varepsilon / \|x_0 - x^*\|) / \log \gamma$ steps. Smaller γ means faster convergence.

4.3 The Bellman optimality operator is a γ -contraction

In MAT31-2 you wrote the Bellman optimality equation as a one-step accounting identity, “value today = best reward today + discounted value tomorrow”, on a small finite example. We now turn that identity into an **operator** and prove that **iterating it converges** — solving the MDP.

Definition 11 A **finite discounted MDP** consists of a finite state space S , finite action sets $A(s)$, a transition kernel $P(s' | s, a)$ (the probability of moving to s' when taking action a in state s , with $\sum_{s'} P(s' | s, a) = 1$), a reward $r(s, a)$, and a **discount factor** $\gamma \in [0, 1)$. A **value function** $V : S \rightarrow \mathbb{R}$ assigns a number to each state; value functions live in $\mathbb{R}^S$ under the **sup norm**

$$\|V\|_\infty = \max_{s \in S} |V(s)|.$$

The **Bellman optimality operator** $T : \mathbb{R}^S \rightarrow \mathbb{R}^S$ is

$$(TV)(s) = \max_{a \in A(s)} \left[r(s, a) + \gamma \sum_{s'} P(s' | s, a) V(s') \right].$$

A value function V^* solving $TV^* = V^*$ is a fixed point of T — exactly the Bellman optimality equation. Banach’s theorem will give it to us, provided T is a contraction.

Proposition 12 (γ -contraction in the sup norm) For any value functions $V, W \in \mathbb{R}^S$,

$$\|TV - TW\|_\infty \leq \gamma \|V - W\|_\infty.$$

Proof. Fix a state s . Using the elementary inequality $|\max_a f(a) - \max_a g(a)| \leq \max_a |f(a) - g(a)|$ (the maxima cannot differ by more than the worst pointwise gap),

$$\begin{aligned} |(TV)(s) - (TW)(s)| &\leq \max_a \left| \gamma \sum_{s'} P(s' | s, a) (V(s') - W(s')) \right| \\ &\leq \max_a \gamma \sum_{s'} P(s' | s, a) |V(s') - W(s')| \\ &\leq \max_a \gamma \sum_{s'} P(s' | s, a) \|V - W\|_\infty = \gamma \|V - W\|_\infty, \end{aligned}$$

where the last step uses $|V(s') - W(s')| \leq \|V - W\|_\infty$ for every s' and then $\sum_{s'} P(s' | s, a) = 1$. The bound holds for every s , so it holds for the maximum over s , which is $\|TV - TW\|_\infty$. $\square$

Theorem 13 (Convergence of value iteration) Since T is a γ -contraction on the complete space $(\mathbb{R}^S, \|\cdot\|_\infty)$, Banach's theorem applies: T has a **unique** fixed point V^* (the **optimal value function**), and for any initial V_0 the **value iteration** $V_{k+1} = TV_k$ converges to V^* with

$$\|V_k - V^*\|_\infty \leq \gamma^k \|V_0 - V^*\|_\infty.$$

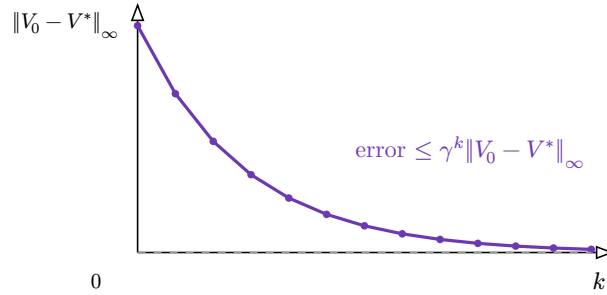


Figure 3: Geometric convergence of value iteration (here $\gamma = 0.7$). Each sweep of the Bellman operator multiplies the worst-case error by at most γ , so the error decays like γ^k — straight-line decay on a logarithmic scale. The discount factor γ is simultaneously the economic “patience” of the agent and the convergence rate of the algorithm.

Common pitfall The contraction factor is the discount γ , and the proof needs $\gamma < 1$ in two places: to make T a contraction at all, and (invisibly) so that the infinite discounted rewards defining V^* are finite. An **undiscounted** problem ($\gamma = 1$) is genuinely harder — T is then only non-expansive, Banach does not apply, and value iteration may fail to converge.

Exercises

13. Show $T(x) = \cos(x)$ is a contraction on $[0, 1]$ and estimate how many iterations bring the error below 10^{-3} from $x_0 = 0$.
14. For a two-state MDP with $\gamma = 0.9$, one action per state, rewards $r(s_1) = 1$, $r(s_2) = 0$, and transitions that stay put, compute V^* by solving $V = TV$ directly, then verify a few steps of value iteration approach it.
15. Prove the inequality $|\max_a f(a) - \max_a g(a)| \leq \max_a |f(a) - g(a)|$ used in the contraction proof.
16. Using the error bound, how many value-iteration sweeps guarantee $\|V_k - V^*\|_\infty \leq 0.01 \|V_0 - V^*\|_\infty$ when $\gamma = 0.95$?

5 Shannon entropy

We now cross into the second half of the course. The organizing question changes from “does this converge?” to “how much **uncertainty** does a random variable carry, and how much does observing one variable **reduce** the uncertainty of another?”. The unit of uncertainty is the **bit**, and the master quantity is **Shannon entropy**.

5.1 From surprise to entropy

Start with a single observation. If an event of probability p occurs, how **surprised** should you be? A near-certain event ($p \approx 1$) is unsurprising; a rare event ($p \approx 0$) is very surprising. Shannon's measure of surprise is

$$\text{surprise}(x) = \log \frac{1}{p(x)} = -\log p(x),$$

which is 0 when $p = 1$ and grows without bound as $p \rightarrow 0$. Using $\log = \log_2$ measures surprise in **bits**: a fair coin landing heads ($p = 1/2$) carries $-\log_2(1/2) = 1$ bit. **Entropy** is just the **average surprise**.

Definition 14 For a discrete random variable X with distribution $p(x)$, the **Shannon entropy** is the expected surprise

$$H(X) = -\sum_x p(x) \log p(x) = \mathbb{E}[-\log p(X)],$$

with the convention $0 \log 0 = 0$ (an impossible outcome contributes nothing). With $\log = \log_2$, entropy is in **bits**.

Worked example — the fair and the biased coin. A fair coin has $H = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = 1$ bit: one yes/no question's worth of uncertainty. A biased coin with $p(\text{heads}) = 0.9$ has $H = -0.9 \log_2 0.9 - 0.1 \log_2 0.1 \approx 0.47$ bits — you are already fairly sure of the outcome, so observing it tells you less. A two-headed coin ($p = 1$) has $H = 0$: no uncertainty, no information in the outcome.

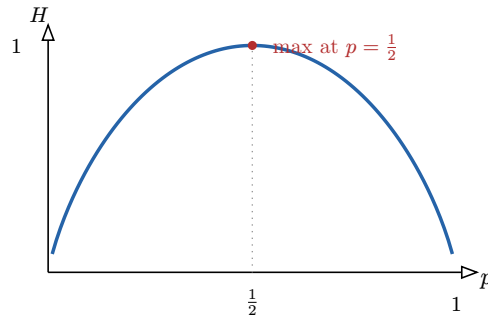


Figure 4: The binary entropy $H(p) = -p \log_2 p - (1-p) \log_2 (1-p)$ of a coin with bias p . It is 0 at the certain outcomes $p = 0$ and $p = 1$, and **maximal** (1 bit) at $p = 1/2$, where the coin is least predictable. Uncertainty peaks at uniformity.

5.2 Basic properties and maximum entropy

Proposition 15 (Bounds on entropy) For a random variable X taking k distinct values:

$$0 \leq H(X) \leq \log k.$$

The lower bound $H(X) = 0$ holds iff X is **deterministic** (one outcome has probability 1).

The upper bound $H(X) = \log k$ holds iff X is **uniform** (every outcome has probability $1/k$).

So entropy ranges from 0 (no uncertainty) to $\log k$ (maximal uncertainty), and **uniformity maximizes entropy** — the most uncertain distribution on a finite alphabet is the one that plays

no favourites. This **maximum entropy principle** is the basis for choosing “least committal” prior distributions.

5.3 Joint, conditional, and the chain rule

With two variables, uncertainty can be measured jointly or conditionally.

Definition 16 For a pair (X, Y) with joint distribution $p(x, y)$:

- the **joint entropy** is $H(X, Y) = -\sum_{x,y} p(x, y) \log p(x, y)$;
- the **conditional entropy** of X given Y is

$$H(X | Y) = -\sum_{x,y} p(x, y) \log p(x | y) = \sum_y p(y) H(X | Y = y),$$

the average, over the values of Y , of the leftover uncertainty in X .

The conditional entropy $H(X | Y)$ answers: **once I have observed Y , how uncertain am I still about X ?** It is a weighted average of the entropies of the conditional distributions $p(x | Y = y)$.

Theorem 17 (Chain rule for entropy)

$$H(X, Y) = H(X) + H(Y | X) = H(Y) + H(X | Y).$$

Proof. Factor the joint $p(x, y) = p(x) p(y | x)$ inside the logarithm:

$$H(X, Y) = -\sum_{x,y} p(x, y) [\log p(x) + \log p(y | x)] = -\sum_x p(x) \log p(x) - \sum_{x,y} p(x, y) \log p(y | x),$$

where the first sum collapses to $H(X)$ (summing $p(x, y)$ over y gives $p(x)$) and the second is $H(Y | X)$. The symmetric factorization gives the other form. $\square$

Remark The chain rule reads exactly as intuition demands: the total uncertainty in the pair equals the uncertainty in X **plus** the uncertainty remaining in Y once X is known. Uncertainty **adds up sequentially**, just like the multiplication rule for probabilities $p(x, y) = p(x)p(y | x)$ from MAT21-2, but in logarithmic (additive) units.

Common pitfall Conditioning never **increases** uncertainty **on average**: $H(X | Y) \leq H(X)$. But this is only true on average — for a **particular** value $Y = y$ it is entirely possible that $H(X | Y = y) > H(X)$, i.e. one specific observation leaves you more confused than before. Do not mistake the average inequality for a guarantee about every individual outcome.

Exercises

17. Compute $H(X)$ in bits for a four-sided die with probabilities $(1/2, 1/4, 1/8, 1/8)$ and compare with the uniform maximum $\log_2 4$.
18. A pair (X, Y) has joint table $p(0, 0) = p(1, 1) = 1/3$, $p(0, 1) = p(1, 0) = 1/6$. Compute $H(X)$, $H(Y)$, $H(X, Y)$, and verify the chain rule.
19. Show that if X and Y are independent then $H(X, Y) = H(X) + H(Y)$.

20. Prove $H(X) \leq \log k$ in the special case $k = 2$ by maximizing the binary entropy $H(p)$ with calculus.

6 Mutual information

Entropy measures uncertainty in **one** variable; the chain rule relates **two**. Subtracting the two gives the single most important quantity connecting them: **mutual information**, the amount by which knowing Y reduces the uncertainty in X . It is the precise measure of “how much Y tells you about X ”.

6.1 Definition: information is reduced uncertainty

Definition 18 The **mutual information** between X and Y is

$$I(X; Y) = H(X) - H(X | Y) = H(Y) - H(Y | X) = \sum_{x,y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}.$$

It is the reduction in uncertainty about X from observing Y (equivalently, about Y from observing X — it is **symmetric**).

The three expressions are equal by the chain rule, and the symmetry is striking: **Y tells you exactly as much about X as X tells you about Y** . The last form, a sum over the joint distribution of $\log[p(x, y)/p(x)p(y)]$, makes the connection to independence transparent — it compares the true joint $p(x, y)$ to what it **would** be if the variables were independent, $p(x)p(y)$.

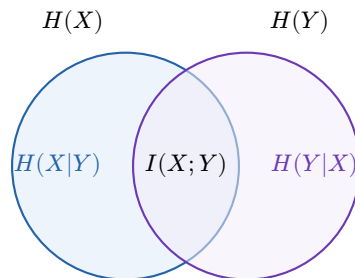


Figure 5: The information diagram. The two circles are the entropies $H(X)$ and $H(Y)$; their **overlap** is the mutual information $I(X; Y)$, the shared uncertainty. The crescents are the conditional entropies $H(X | Y)$ and $H(Y | X)$ — what is left of each variable once the other is known. The whole figure is the joint entropy $H(X, Y) = H(X) + H(Y) - I(X; Y)$.

6.2 Mutual information and independence

Theorem 19 (Information detects independence) $I(X; Y) \geq 0$ always, with $I(X; Y) = 0$ if and only if X and Y are independent.

Proof. The defining sum is exactly the KL divergence (Chapter 10) of the true joint $p(x, y)$ from the product of marginals $p(x)p(y)$:

$$I(X; Y) = D_{\text{KL}}(p(x, y) \parallel p(x)p(y)).$$

KL divergence is always ≥ 0 and equals 0 iff its two arguments coincide (Gibbs’ inequality, proved in Chapter 10). Here that means $p(x, y) = p(x)p(y)$ for all x, y — precisely the definition of independence from MAT21-2. $\square$

This is a beautiful unification: the probabilistic notion of **independence** (“knowing Y tells you nothing about X ”) and the information-theoretic notion (“ $I(X; Y) = 0$ ”) are literally the same statement.

6.3 The data processing inequality

A final structural law: **you cannot create information about X by processing Y .**

Theorem 20 (Data processing inequality) If $X \rightarrow Y \rightarrow Z$ is a **Markov chain** — meaning Z depends on X only through Y , i.e. $p(z \mid x, y) = p(z \mid y)$ — then

$$I(X; Z) \leq I(X; Y).$$

Remark The slogan is “**post-processing cannot increase information**”. Whatever transformation $Z = g(Y)$ you apply to your data — averaging, quantizing, running it through a neural network layer — the result Z can carry **at most** as much information about the underlying X as the raw Y did. Information about the source can be preserved or destroyed, never manufactured. In machine learning this is a caution: no feature engineering on Y alone recovers information about the label X that Y did not already contain.

Common pitfall The inequality requires the **Markov** structure $X \rightarrow Y \rightarrow Z$. If Z has its own independent access to X (so it does **not** factor through Y), then of course Z can carry more information about X than Y does — the chain is broken and the theorem does not apply. Always check that Z sees X only via Y .

Exercises

21. For the joint table $p(0, 0) = p(1, 1) = 1/3$, $p(0, 1) = p(1, 0) = 1/6$ (Chapter 5), compute $I(X; Y)$ and confirm it is positive, so X and Y are dependent.
22. Show that $I(X; X) = H(X)$: a variable shares all of its information with itself.
23. Verify $I(X; Y) = 0$ for an independent pair, e.g. two independent fair coins.
24. Let Y be a noisy copy of X and $Z = g(Y)$ a deterministic function. Argue from the data processing inequality that Z cannot be a better predictor of X than Y in the information sense.

7 Exponential families

We turn from describing distributions to **fitting** them to data. A remarkably large and important class of models — Bernoulli, binomial, Poisson, normal, exponential, gamma — share a common algebraic skeleton, the **exponential family**. Writing a distribution in this canonical form exposes its **natural parameter** and **sufficient statistic**, and reduces maximum likelihood estimation to a single clean equation. This is the structure that makes the MLE computations of MAT21-2 systematic rather than ad hoc.

7.1 Canonical form

Definition 21 A family of distributions $p(x; \eta)$ is an **exponential family in canonical (natural) form** if

$$p(x; \eta) = h(x) \exp(\eta T(x) - A(\eta)),$$

where

- η is the **natural parameter**,
- $T(x)$ is the **sufficient statistic**,
- $h(x)$ is the **base measure** (carries the part not depending on η),
- $A(\eta) = \log \sum_x h(x) \exp(\eta T(x))$ is the **log-partition function**, the normalizer (an integral replaces the sum in the continuous case).

The log-partition function $A(\eta)$ is not a free choice — it is forced, being exactly what makes $p(x; \eta)$ sum to 1. As we will see, it secretly stores all the moments of $T(X)$.

7.2 Recognizing the structure

Worked example — Bernoulli in canonical form. For $X \sim \text{Bernoulli}(p)$, the mass function $p^x(1-p)^{1-x}$ does not **look** like the canonical form — but take its logarithm and reassemble:

$$p^x(1-p)^{1-x} = \exp(x \log p + (1-x) \log(1-p)) = \exp\left(x \log \frac{p}{1-p} + \log(1-p)\right).$$

Matching to $\exp(\eta T(x) - A(\eta))$ with $h(x) = 1$:

$$T(x) = x, \quad \eta = \log \frac{p}{1-p} \quad (\text{the log-odds}), \quad A(\eta) = -\log(1-p) = \log(1 + e^\eta).$$

The natural parameter of a Bernoulli is its **log-odds** — exactly the quantity a logistic-regression model is linear in.

Worked example — Poisson in canonical form. For $X \sim \text{Poisson}(\lambda)$, $p(x) = e^{-\lambda} \lambda^x / x!$. Write

$$e^{-\lambda} \frac{\lambda^x}{x!} = \frac{1}{x!} \exp(x \log \lambda - \lambda).$$

So $h(x) = 1/x!$, $T(x) = x$, the natural parameter is $\eta = \log \lambda$, and $A(\eta) = \lambda = e^\eta$.

7.3 The log-partition function generates moments, and the MLE

Here is the payoff that makes the exponential-family viewpoint so powerful: differentiating A produces the mean and variance of the sufficient statistic.

Theorem 22 (Moments from the log-partition function)

$$A'(\eta) = \mathbb{E}_\eta[T(X)], \quad A''(\eta) = \text{Var}_\eta(T(X)).$$

In particular A is **convex** (its second derivative, a variance, is ≥ 0).

Proof. Differentiate $A(\eta) = \log \sum_x h(x) e^{\eta T(x)}$. Writing $Z(\eta) = \sum_x h(x) e^{\eta T(x)} = e^{A(\eta)}$,

$$A'(\eta) = \frac{Z'(\eta)}{Z(\eta)} = \frac{\sum_x T(x) h(x) e^{\eta T(x)}}{Z(\eta)} = \sum_x T(x) p(x; \eta) = \mathbb{E}_{\eta[T(X)]}.$$

Differentiating once more gives $A''(\eta) = \mathbb{E}[T^2] - (\mathbb{E}[T])^2 = \text{Var}_{\eta(T(X))} \geq 0$. $\square$

Theorem 23 (MLE by moment matching) Given i.i.d. data $x_1, \dots, x_n$ from an exponential family, with sample mean of the sufficient statistic $\bar{T} = \frac{1}{n} \sum_{i=1}^n T(x_i)$, the maximum-likelihood estimate $\hat{\eta}$ is the solution of

$$A'(\hat{\eta}) = \bar{T}, \quad \text{equivalently} \quad \mathbb{E}_{\hat{\eta}}[T(X)] = \bar{T}.$$

“Choose the parameter whose predicted mean of T equals the observed mean of T .”

Proof. The log-likelihood is $\ell(\eta) = \sum_i (\eta T(x_i) - A(\eta)) + \text{const} = \eta \sum_i T(x_i) - nA(\eta) + \text{const}$. Setting $\ell'(\eta) = \sum_i T(x_i) - nA'(\eta) = 0$ gives $A'(\hat{\eta}) = \frac{1}{n} \sum_i T(x_i) = \bar{T}$. Because A is convex, ℓ is concave, so this stationary point is the unique global maximum. $\square$

Worked example — recovering the Bernoulli MLE. For the Bernoulli, $T(x) = x$ and $A'(\eta) = \mathbb{E}[X] = p$. The moment-matching equation reads $p = \bar{x} = \frac{1}{n} \sum_i x_i$ — the MLE of the success probability is the sample frequency, exactly the result you derived directly in MAT21-2, now falling out of a single general principle.

Common pitfall Sign and placement of $A(\eta)$ are the usual stumbling block: it sits **inside** the exponential as $-A(\eta)$, and it depends on η **only**, never on x . Anything that depends on x but not η belongs in $h(x)$; anything depending on both must be reshaped until it splits as $\eta T(x)$. If you cannot achieve that split, the distribution is not in the (single-parameter) exponential family.

Exercises

25. Write the exponential distribution with rate λ , density $\lambda e^{-\lambda x}$ on $x > 0$, in canonical form: identify T , η , h , and $A(\eta)$.
26. For the Poisson family, compute $A'(\eta)$ and $A''(\eta)$ and verify they give mean λ and variance λ .
27. Derive the Poisson MLE $\hat{\lambda} = \bar{x}$ from the moment-matching equation.
28. Show that the normal distribution with known variance σ^2 is an exponential family with $T(x) = x$ and natural parameter $\eta = \mu/\sigma^2$.

8 Fisher information

How much does a **single data point** tell you about an unknown parameter θ ? The answer is the **Fisher information** $I(\theta)$ — a number that quantifies how sharply the data pin down the parameter, and that will, in the next chapter, set the ultimate limit on estimation accuracy. It is built from the **score**, the sensitivity of the log-likelihood to the parameter.

8.1 The score function

Definition 24 For a parametric model with log-likelihood $\ell(\theta; x) = \log p(x; \theta)$, the **score** is its derivative in the parameter,

$$s(\theta; x) = \frac{\partial}{\partial \theta} \log p(x; \theta) = \frac{\partial_{\theta} p(x; \theta)}{p(x; \theta)}.$$

The score measures, for a given observation x , **how strongly the likelihood prefers a slightly larger θ** . A key regularity fact is that, averaged over the data, the model is unbiased about the truth:

Proposition 25 (Mean-zero score) Under regularity conditions allowing differentiation under the integral sign,

$$\mathbb{E}_\theta[s(\theta; X)] = 0.$$

Proof. Differentiate the identity $\int p(x; \theta) dx = 1$ in θ and swap derivative and integral (an **interchange theorem** from Chapter 3 in action):

$$0 = \partial_\theta \int p(x; \theta) dx = \int \partial_\theta p(x; \theta) dx = \int s(\theta; x) p(x; \theta) dx = \mathbb{E}_{\theta[s(\theta; X)]}.$$

□

8.2 Two faces of Fisher information

Definition 26 The **Fisher information** of the model at θ , for one observation, is the variance of the score:

$$I(\theta) = \text{Var}_{\theta(s(\theta; X))} = \mathbb{E}_{\theta[s(\theta; X)^2]}$$

(the second equality holds because the score has mean zero).

Theorem 27 (Variance-of-score = negative expected Hessian) Under the same regularity conditions,

$$I(\theta) = \mathbb{E}_{\theta[s(\theta; X)^2]} = -\mathbb{E}_\theta\left[\frac{\partial^2}{\partial \theta^2} \log p(X; \theta)\right].$$

Proof. Differentiate $\mathbb{E}_{\theta[s]} = 0$, i.e. $\int (\partial_\theta \log p) p dx = 0$, once more in θ . By the product rule inside the integral,

$$\int (\partial_\theta^2 \log p) p dx + \int (\partial_\theta \log p)(\partial_\theta p) dx = 0.$$

In the second integral write $\partial_\theta p = (\partial_\theta \log p) p$, so it becomes $\int (\partial_\theta \log p)^2 p dx = \mathbb{E}[s^2]$. Hence $\mathbb{E}[\partial_\theta^2 \log p] + \mathbb{E}[s^2] = 0$, which rearranges to the claim. □

These are the two definitions the OP names. The first, $\mathbb{E}[s^2]$, says information is the **expected squared sensitivity** of the likelihood. The second, $-\mathbb{E}[\partial_\theta^2 \log p]$, says information is the **expected curvature** of the log-likelihood — and that is the picture to carry in your head.

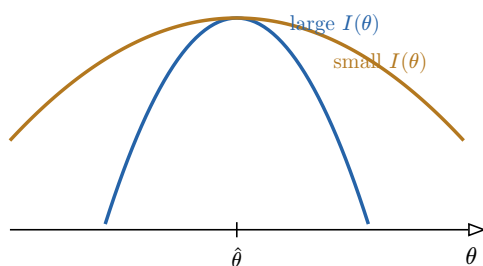


Figure 6: Fisher information as curvature of the log-likelihood at its peak $\hat{\theta}$. A **sharply** peaked log-likelihood (blue, large $I(\theta)$) pins the parameter down precisely — a small move in θ costs a lot of likelihood. A **flat** peak (orange, small $I(\theta)$) leaves the parameter poorly determined. Curvature **is** information.

Remark (additivity over data) For n independent observations the log-likelihood is a **sum**, so its curvature adds: the total Fisher information is $n I(\theta)$. More data means proportionally more information and, as the next chapter shows, proportionally smaller achievable variance.

Common pitfall Fisher information is a property of the **model and the true θ** , computed by taking an **expectation over the data** — it is not a function of any particular sample. Do not confuse it with the **observed** second derivative at one dataset (the “observed information”); they coincide in expectation but the Fisher information is the averaged, population quantity.

Exercises

29. For the Bernoulli(p) model compute the score $s(p; x)$ and show $I(p) = 1/(p(1 - p))$ — minimal at $p = 1/2$ and exploding near the endpoints. Interpret.
30. For the Poisson(λ) model show $I(\lambda) = 1/\lambda$.
31. For the normal $N(\mu, \sigma^2)$ with known σ^2 , show the Fisher information about μ is $1/\sigma^2$.
32. Verify the two definitions agree for the Bernoulli model by computing both $\mathbb{E}[s^2]$ and $-\mathbb{E}[\partial_p^2 \log p]$.

9 The Cramér–Rao lower bound

Fisher information is not merely interpretive: it sets a **hard floor** on how well any unbiased estimator can possibly do. No cleverness, no algorithm, can beat it. This is the Cramér–Rao bound, and it turns “how much information is in the data” into “how accurately can I possibly estimate θ ”.

9.1 The bound

Theorem 28 (Cramér–Rao lower bound) Let $\hat{\theta}$ be any **unbiased** estimator of θ based on n i.i.d. observations (so $\mathbb{E}[\hat{\theta}] = \theta$). Then

$$\text{Var}(\hat{\theta}) \geq \frac{1}{n I(\theta)},$$

where $I(\theta)$ is the Fisher information of a single observation. No unbiased estimator can have smaller variance.

The bound is intuitive in the light of the previous chapter: $nI(\theta)$ is the total information in the sample, and the best possible variance is its reciprocal. **More information $\Rightarrow$ smaller achievable variance.** The floor drops like $1/n$ as data accumulate.

9.2 Efficiency

Definition 29 An unbiased estimator is **efficient** if it **attains** the Cramér–Rao bound:

$$\text{Var}(\hat{\theta}) = \frac{1}{nI(\theta)}.$$

An efficient estimator is the best possible unbiased estimator — it squeezes every drop of information out of the data.

Worked example — the sample mean is efficient for the normal mean. For $N(\mu, \sigma^2)$ with known σ^2 , the single-observation information is $I(\mu) = 1/\sigma^2$, so the bound is $\text{Var}(\hat{\mu}) \geq \sigma^2/n$. The natural estimator $\hat{\mu} = \bar{X}$ has variance exactly σ^2/n . It **attains** the bound — the sample mean is an **efficient** estimator of the normal mean. You cannot do better with any unbiased estimator.

9.3 Minimum sample size

Remark (how much data do I need?) Rearranging the bound answers a practical engineering question. To guarantee that **some** unbiased estimator can achieve variance at most v , you need $1/(nI(\theta)) \leq v$, i.e.

$$n \geq \frac{1}{vI(\theta)}.$$

Half the target variance, twice the data; a model with twice the Fisher information needs half the data for the same precision. This is the quantitative core of **experiment design**.

Common pitfall The Cramér–Rao bound is a statement about **unbiased** estimators only. **Biased** estimators can — and routinely do — achieve **lower** variance than the bound; the whole field of regularization (ridge regression, shrinkage) deliberately accepts a little bias to buy a large reduction in variance. The bound floors variance only after you have committed to unbiasedness.

Exercises

33. For the Bernoulli model with $I(p) = 1/(p(1-p))$, state the Cramér–Rao bound and verify that $\hat{p} = \bar{X}$ attains it.

34. Using $I(\lambda) = 1/\lambda$ for the Poisson, how many observations are needed so an unbiased estimator of $\lambda \approx 4$ can reach variance ≤ 0.01 ?

35. Explain why an estimator with $\text{Var}(\hat{\theta}) = 1/(2nI(\theta))$ would be impossible if it were unbiased, and what its existence would imply about bias.
36. For the normal mean, confirm the sample mean's variance equals the bound and is therefore efficient.

10 Kullback–Leibler divergence

Mutual information was secretly a special case of a more basic quantity: a measure of how **far apart** two distributions are. The **Kullback–Leibler divergence** is that measure — the “extra surprise” you incur by modelling data drawn from P using the wrong distribution Q . It is the bridge from information theory to the loss functions of machine learning.

10.1 Definition and the non-negativity that powers everything

Definition 30 The **Kullback–Leibler (KL) divergence** from Q to P is

$$D_{\text{KL}}(P \parallel Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)} = \mathbb{E}_{X \sim P} \left[\log \frac{P(X)}{Q(X)} \right]$$

(an integral in the continuous case). The sum is taken over x with $P(x) > 0$.

Read it as: **the expected log-likelihood-ratio between the truth P and the model Q , averaged under the truth**. It measures the penalty for using Q when reality is P .

Theorem 31 (Gibbs' inequality)

$$D_{\text{KL}}(P \parallel Q) \geq 0, \quad \text{with equality if and only if } P = Q.$$

Proof. Use the inequality $\log t \leq t - 1$ (with equality only at $t = 1$), applied to $t = Q(x)/P(x)$:

$$-D_{\text{KL}}(P \parallel Q) = \sum_x P(x) \log \frac{Q(x)}{P(x)} \leq \sum_x P(x) \left(\frac{Q(x)}{P(x)} - 1 \right) = \sum_x Q(x) - \sum_x P(x) = 1 - 1 = 0.$$

So $D_{\text{KL}}(P \parallel Q) \geq 0$. Equality requires $Q(x)/P(x) = 1$ for every x with $P(x) > 0$, i.e. $P = Q$. $\square$

This single inequality is the engine behind the whole second half: it is why mutual information is non-negative (Chapter 6), why entropy is maximized by the uniform distribution (Chapter 5), and — next chapter — why cross-entropy is minimized at the truth.

10.2 Asymmetry and pathologies

Common pitfall KL divergence is **not symmetric**: in general $D_{\text{KL}}(P \parallel Q) \neq D_{\text{KL}}(Q \parallel P)$, and it does **not** satisfy the triangle inequality. It is therefore **not a distance**, despite measuring “how far Q is from P ”. Always keep track of which argument is the **truth** (the first, P , under which the expectation is taken) and which is the **model** (the second, Q).

Worked example — the two directions differ. Let $P = (1/2, 1/2)$ and $Q = (1/4, 3/4)$ on two outcomes. Then

$$D_{\text{KL}}(P \parallel Q) = \frac{1}{2} \log \frac{\frac{1}{2}}{\frac{1}{4}} + \frac{1}{2} \log \frac{\frac{1}{2}}{\frac{2}{3}} = \frac{1}{2} \log 2 + \frac{1}{2} \log \frac{2}{3} \approx 0.21 \text{ bits},$$

whereas computing $D_{\text{KL}}(Q \parallel P)$ with the roles swapped gives a **different** number (≈ 0.19 bits). Same pair of distributions, two different divergences.

Common pitfall (the infinite-divergence trap) KL divergence blows up when the **supports** disagree. If $Q(x) = 0$ at some x where $P(x) > 0$, the term $P(x) \log[P(x)/0] = +\infty$, so $D_{\text{KL}}(P \parallel Q) = +\infty$. Intuitively, a model Q that declares an event **impossible** when it actually occurs is infinitely surprised. This is why models in practice keep all probabilities strictly positive (smoothing): a single hard zero in the wrong place makes the loss infinite.

Exercises

37. Compute both $D_{\text{KL}}(P \parallel Q)$ and $D_{\text{KL}}(Q \parallel P)$ for $P = (0.9, 0.1)$ and $Q = (0.5, 0.5)$, confirming asymmetry.
38. Show that for any P , $D_{\text{KL}}(P \parallel U) = \log k - H(P)$ where U is uniform on k outcomes; deduce that KL-to-uniform is minimized exactly when P is uniform.
39. Construct P, Q on three outcomes with $D_{\text{KL}}(P \parallel Q) = +\infty$ but $D_{\text{KL}}(Q \parallel P) < \infty$.
40. Verify Gibbs' inequality numerically on a pair of your choice and check that equality requires $P = Q$.

11 Cross-entropy and the link to learning

Everything in the second half converges on a single, practical destination: the **cross-entropy loss** that trains essentially every modern classifier. We now show that minimizing it is **identical** to maximum likelihood, tracing the argument cleanly from KL divergence through cross-entropy to the loss function — and thereby connecting this entire course to neural-network training.

11.1 Cross-entropy and its decomposition

Definition 32 The **cross-entropy** of the model Q relative to the truth P is

$$H(P, Q) = - \sum_x P(x) \log Q(x) = \mathbb{E}_{X \sim P}[-\log Q(X)].$$

It is the **average surprise** of an observer who believes the distribution is Q while data actually arrive from P .

When $Q = P$ this is just the entropy $H(P)$ — the irreducible average surprise. When $Q \neq P$ the surprise is **larger**, and the excess is exactly the KL divergence.

Proposition 33 (Cross-entropy = entropy + divergence)

$$H(P, Q) = H(P) + D_{\text{KL}}(P \parallel Q).$$

Proof.

$$H(P, Q) - H(P) = - \sum_x P(x) \log Q(x) + \sum_x P(x) \log P(x) = \sum_x P(x) \log \frac{P(x)}{Q(x)} = D_{\text{KL}}(P \parallel Q).$$

□

This is the relation the OP requires, and it is the conceptual hinge of the chapter. Since $D_{\text{KL}}(P \parallel Q) \geq 0$ with equality iff $Q = P$ (Gibbs), the cross-entropy $H(P, Q)$ is **minimized over Q exactly when $Q = P$** , and its minimum value is $H(P)$. **The truth is the unique minimizer of cross-entropy.**

11.2 Why minimizing cross-entropy is maximizing likelihood

Theorem 34 (Cross-entropy minimization = maximum likelihood) Let P be the **empirical distribution** of observed data $x_1, \dots, x_n$ (mass $1/n$ on each datum) and let Q_θ be a parametric model. Then the cross-entropy is the **negative average log-likelihood**:

$$H(P, Q_\theta) = -\frac{1}{n} \sum_{i=1}^n \log Q_\theta(x_i) = -\frac{1}{n} \ell(\theta).$$

Hence **minimizing cross-entropy over θ is exactly maximizing the log-likelihood $\ell(\theta)$.**

Proof. With P the empirical distribution, $P(x) = \frac{1}{n} \times (\text{count of } x)$, so

$$H(P, Q_\theta) = -\sum_x P(x) \log Q_\theta(x) = -\frac{1}{n} \sum_{i=1}^n \log Q_\theta(x_i),$$

which is $-1/n$ times the log-likelihood $\ell(\theta) = \sum_i \log Q_\theta(x_i)$. Minimizing the left side and maximizing $\ell(\theta)$ are the same optimization. □

Remark (the neural-network picture) In classification, a network with parameters θ outputs a predicted distribution $Q_{\theta(\cdot \mid \text{input})}$ over the classes (the softmax layer), and the target P is the **one-hot** empirical distribution placing all mass on the true label. The training loss

$$H(P, Q_\theta) = -\log Q_{\theta(\text{true label} \mid \text{input})}$$

is precisely the cross-entropy. By the decomposition $H(P, Q_\theta) = H(P) + D_{\text{KL}}(P \parallel Q_\theta)$ and $H(P) = 0$ for a one-hot P , minimizing the loss drives $D_{\text{KL}}(P \parallel Q_\theta) \rightarrow 0$, i.e. pushes the predicted distribution toward the true label — which, by the theorem, is identical to maximizing the likelihood of the data. **Cross-entropy training and maximum likelihood are one and the same procedure.**

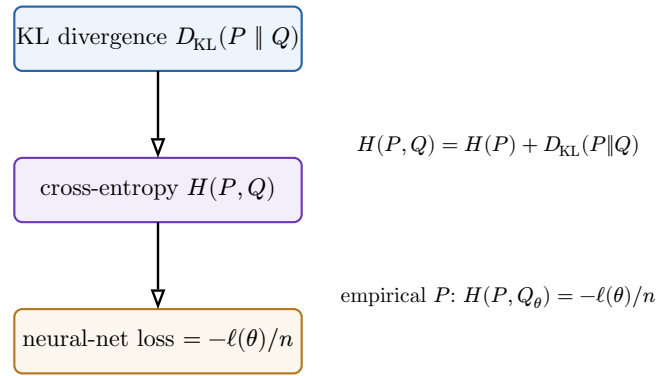


Figure 7: The argument of the final chapter, top to bottom. KL divergence measures model error; adding the constant $H(P)$ turns it into cross-entropy; taking P to be the empirical distribution turns cross-entropy into the negative average log-likelihood — the loss a network minimizes. Each arrow is an identity proved in the text, so minimizing the bottom box maximizes likelihood.

Common pitfall The equivalence relies on $H(P)$ being **independent of the parameters** θ — which it is, because the empirical distribution P depends only on the data, not on the model. This is what lets us drop $H(P)$ and minimize $D_{\text{KL}}(P \parallel Q_\theta)$ and the cross-entropy interchangeably. If you ever parameterize P itself, this clean separation breaks.

Exercises

41. For a single binary label with target $P = (1, 0)$ and model prediction $Q_\theta = (q, 1 - q)$, write the cross-entropy loss and show it equals $-\log q$.
42. Using the decomposition, explain why the minimum achievable cross-entropy on data with true distribution P is $H(P)$, not 0, unless P is one-hot.
43. Show that for the empirical distribution the cross-entropy loss averaged over a dataset is $-1/n$ times the log-likelihood, for a Bernoulli model $Q_\theta = \text{Bernoulli}(\theta)$, and recover $\hat{\theta} = \bar{x}$.
44. Explain, citing Gibbs' inequality, why no model Q can drive the cross-entropy $H(P, Q)$ below the entropy $H(P)$.